

Alexander Bernstein

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Education

University of California, Santa Barbara

Ph.D. in Statistics

Santa Barbara, CA

Sept. 2025

Dissertation: “Long-Only Minimum Variance Portfolio Composition for Factor Models”

Washington University in St. Louis

M.Sc. in Systems Science and Applied Mathematics

St. Louis, MO

Dec. 2016

B.A. in Mathematics and Economics (Cum Laude)

May 2014

Technical Skills

Languages & Tools: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), SQL, R, C, Java, JavaScript/NodeJS, Git, Linux, AWS, MongoDB, $\text{\LaTeX}$.

Machine Learning: Neural Networks, Gaussian Processes, Reinforcement Learning (Q-Learning), Ensemble Methods (Boosting/Bagging), Bayesian Inference, Time Series Forecasting, Feature Engineering.

Data & Modeling: Statistical Modeling, Experimentation, High-Dimensional Regression, Optimization, Backtesting & Simulation, Data Pipelines.

Selected Projects

End-to-End Equity Modeling Pipeline: Architected a modular, spec-driven (**YAML**) **Python** pipeline for S&P 500 portfolio construction: pluggable covariance/return estimators via decorator registries, **Numba**-JIT Kalman filters, **CVXPY** optimization, a walk-forward backtest engine, and a CLI rebalancer that emits live trades. Built on clean ABCs with unit tests and reproducible configs; runs the full **S&P 500** (daily/weekly data since 2010) in ~ 5 – 10 min per strategy.

ML Surrogates for Derivative Pricing:  Trained **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates to emulate Monte Carlo and finite-difference (Crank-Nicolson) solvers for the **Heston** and **CEV** models, recovering prices and **Greeks** (via autograd) at a fraction of the numerical cost.

Conditional Covariance Estimation:  Built a **Covariance Regression** model (**CovRegpy**) for covariate-adjusted covariance of idiosyncratic equity returns — a scalable alternative to **MGARCH** that sharply reduces parameter count in high dimensions.

Multi-Agent Reinforcement Learning:  Implemented a **Q-Learning** agent for a mean-field cyber-security environment, learning robust defense policies against a non-stationary population distribution μ .

Experience

University of California, Santa Barbara

Teaching Assistant & Graduate Student Mentor

Santa Barbara, CA

Sept. 2017 – June 2025

Taught statistics, probability, regression, time series, and stochastic processes; mentored undergraduate research projects in modeling and optimization, including posters and written reports.

Epic Systems

Technical Services Engineer

Madison, WI

Sept. 2014 – Sept. 2015

Diagnosed customer issues and shipped improvements to the Epic EHR codebase for large institutional clients under high-availability and regulatory constraints.

Prozess Technologie

Intern

St. Louis, MO

May 2013 – May 2014

Built computational simulations to validate laboratory equipment and calibrated pharmaceutical manufacturing hardware to industry standards.

Research & Publications

Banerjee, T., **Bernstein, A.**, Feinstein, Z. (2025). “Dynamic clearing and contagion in financial networks.” *European Journal of Operational Research* 321(2), 664–675.

Bernstein, A., Shkolnik, A. (2025). “Asymptotics of Quadratic Forms on a Simplex.” *In preparation*.